

LAB-3 Object and class

1. What is Object-Oriented Programming (OOP)? Mention any four benefits of OOP.
2. Define class and object. Explain the relationship between them with a simple real-world example.
3. What is meant by class instance in Java? How is an object created?
4. Explain the structure of a simple Java class. What are the roles of:
 - a) data members
 - b) methods
 - c) constructors
5. What is abstraction? How is abstraction achieved in Java?
6. What is encapsulation? Explain how encapsulation improves data security with an example idea.
7. Differentiate between abstraction and encapsulation with at least 3 points.
8. Explain the use of the this keyword in Java. Write a short code snippet where this is necessary.
9. What is a constructor? List the characteristics of constructors in Java. Differentiate between:
 - a) default constructor
 - b) parameterized constructor
10. What is meant by access control in Java? Explain the four access specifiers (private, default, protected, public).
11. Explain method overloading. What rules must be followed for valid overloading? Differentiate between:
 - a) method overloading and method overriding.
12. Java is said to be pass-by-value. Explain this statement with a clear example involving objects.
13. What are nested classes and inner classes in Java? Differentiate between them and mention at least two uses.

Programming problems

14. Write a program to implement encapsulation in a class Person using private variables and public getters/setters.
15. Write a program to create a class Car where:
 - a) speed is private
 - b) methods accelerate() and brake() update speed
 - c) speed cannot go below 0
16. Write a program to demonstrate method overloading by creating multiple add() methods for:
 - a) two integers
 - b) three integers
 - c) two doubles
17. Write a program to create a class Calculator with methods returning values:
 - a) sum(int a, int b)
 - b) product(int a, int b)
 - c) max(int a, int b)
18. Write a program to show constructor overloading in a class Book:
 - a) default constructor
 - b) parameterized constructor (title, author, price)

- c) Write a Java program to demonstrate compile-time polymorphism using method overloading in a real scenario (example: printBill() in a shop).
- 19. Write a recursive Java program to find: factorial of a number (must use recursion only)
- 20. Write a recursive Java program to find: nth Fibonacci number (must use recursion only)
- 21. Write a Java program to demonstrate an inner class:
 - a) Outer class: University
 - b) Inner class: Department
 - c) Print department details using inner class object.
- 22. Write a Java program to demonstrate a nested static class:
 - a) Outer class: Computer
 - b) Static nested class: CPU
 - c) Show how objects are created and accessed.